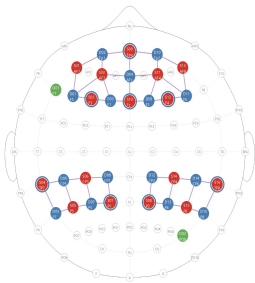


DATA

fNIRS Group

N = 21
42-channel
Collect data on
portable cap



Different participants

fMRI Group

N = 59
122 ROIs
Whole-brain
Open dataset

STIMULUS

North by Northwest

~15 min Hitchcock clip
886 timepoints
TR = 1s

Temporal Alignment

Trim raw recordings
Align clips by the start and
end markers

Cognitive Tasks

The same N = 21 from fNIRS group

1. **N-back** (1- and 2-back)
Working memory recall
2. **CPT** (sustained attention)
Go/No-go vigilance
3. **CPT-MEM**
Attention + Memory

MODEL TRAINING

aPCR Model

(Gao et al., 2025)

1. **PCA**: fNIRS (~25 PCs) & fMRI (~66 PCs)
2. **OLS**: fNIRS PCs → fMRI PCs
3. **LOO cross-validation** (21 folds)

Trained weights

Prediction (movie)

Predicted Whole-Brain fMRI from
movie fNIRS

37/122 ROIs significant (FDR $q < 0.05$)
Inter-Subject FC: $r = 0.552$

CROSS-TASK TRANSFER

FREEZE MODEL

No retraining, predict whole-brain
fMRI from task fNIRS

Planned Validation

1. DAN/CONT activation ↑ and DMN ↓ from 1-back to 2-back, matching known fMRI load effects
2. Higher predicted DMN activation → more variable reaction times (attentional lapses)
3. Higher predicted DAN/CONT activation → better accuracy and CPT d-prime
4. Prediction R^2 higher for 2-back than 1-back, confirming sensitivity to cognitive load